

Lab 3 Quiz: IMU Data Analysis

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Quiz Overview

This quiz covers IMU sensor data analysis including gyroscope readings, accelerometer measurements, orientation data, and Allan variance characterization for the VectorNav VN-100 IMU sensor.

Question 1 (1 pt)

Plot time series of gyro x, y, z on one plot (Fig 0)

Answer:

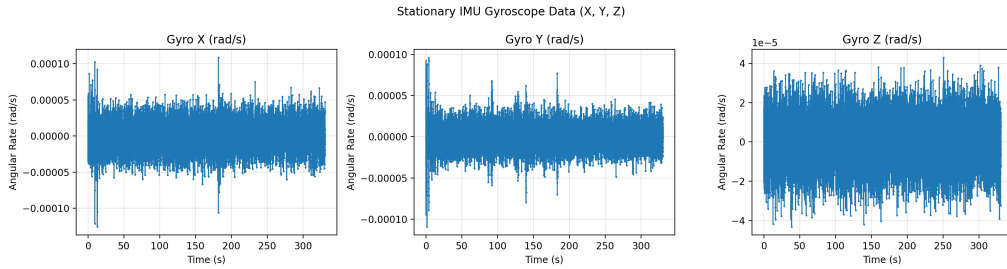


Figure 1: Time series plot of gyroscope data (x, y, z axes) for stationary IMU

Question 2 (1 pt)

Plot accel x, y, z on a second plot (Fig 1)

Answer:

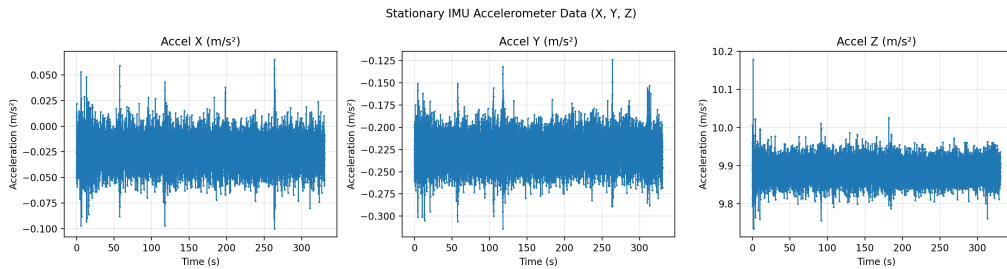


Figure 2: Time series plot of accelerometer data (x, y, z axes) for stationary IMU

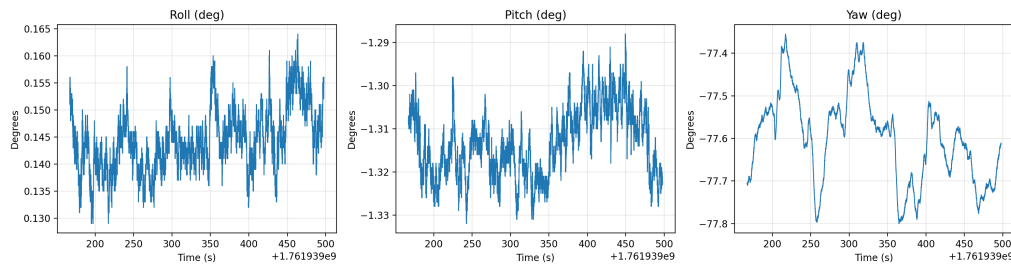
Question 3 (1 pt)**Plot orientation x, y, z (Fig 2)****Answer:**

Figure 3: Time series plot of orientation data (roll, pitch, yaw) for stationary IMU

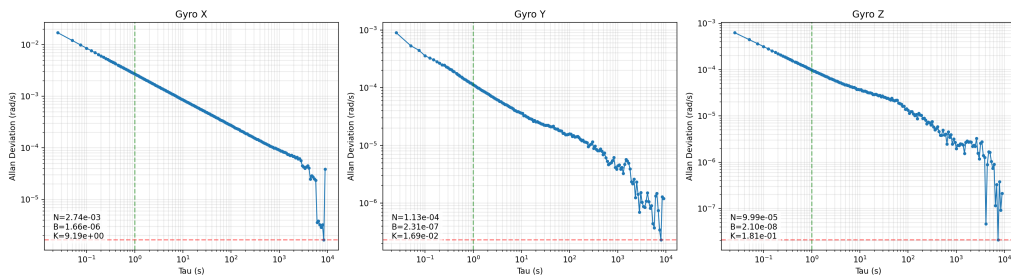
Question 4 (1 pt)**Make Allan variance plots for gyro x, y, z (Fig 3)****Answer:**

Figure 4: Allan variance plots for gyroscope x, y, z axes

Question 5 (1 pt)

What were your parameters for K, B, and N from your Allan variance plot? How did you determine them?

Answer:

For our X axis, we took the values of K, B and N as 9.19×10^{-0} , 1.66×10^{-06} and 2.74×10^{-3}

For our Y axis, we took the values of K, B and N as 1.69×10^{-4} , 2.31×10^{-7} and 1.13×10^{-4}

For our Z axis, we took the values of K, B and N as 1.81×10^{-1} , 2.10×10^{-9} and 9.99×10^{-5}

We determined the value of K by taking tau as 1s (marked by the green line on the plot), the value of B as the minimum point of the allan deviation curve (given by the red line on the plot) and the value of K for the part of the plot where the curve slopes upwards at around +0.5.

Question 6 (1 pt)

Use the time series data from all axes to align the video time with the VectorNav data. Plot three different short "clips" of 1-5 seconds of interesting/relevant IMU

output (doesn't need to be all sensors or axes, Fig 6-8) and include the three screen-grabs near where in the video that the identified motion occurred to illustrate.

Answer:

Analysis with plots and video available at:

<https://1drv.ms/p/c/a9aafc70597a8474/IQD0fqWZSD-HT7ICxvhLisb6AcBkpUfbJTPjABVJWH2rQfY?e=jagcTX>

Question 7 (1 pt)

Use the time series data from all axes to align the video time with the VectorNav data. Plot three different short "clips" of 1-5 seconds of interesting/relevant IMU output (doesn't need to be all sensors or axes, Fig 6-8) and include the three screen-grabs near where in the video that the identified motion occurred to illustrate.

Answer:

Analysis with plots and video available at:

<https://1drv.ms/p/c/a9aafc70597a8474/IQDSQB7-f03sT40HH6xhAkBsAePnJLkCyUoIKQiTgRIqVvk?e=bTfo0l>

Question 8 (1 pt)

Use the time series data from all axes to align the video time with the VectorNav data. Plot three different short "clips" of 1-5 seconds of interesting/relevant IMU output (doesn't need to be all sensors or axes, Fig 6-8) and include the three screen-grabs near where in the video that the identified motion occurred to illustrate.

Answer:

Analysis with plots and video available at:

<https://1drv.ms/p/c/a9aafc70597a8474/IQAbfWRGZqwSTaQZbLjBR8VtAbJclPweVjxhTNUSAsc7Ro?e=Kku5tk>

Question 9 (1 pt)

Describe your experimental setup and motion recording. What trends did you observe in your data during the motion? Are these aligned with what you expect given your understanding of IMUs and what you also observed from the video? (3-4 sentences).

Answer:

The experimental setup included plugging in the VectorNav unit to the PC, running the driver, and recording the published data /imu to a rosbag.

The choice of movement we picked for the IMU was to move it upwards and downwards in a repeated fashion, for 11 seconds and performing an analysis on gyro, acceleration and rotational rate using a CSV file. (We converted the rosbag to CSV for better analysis)

As per my analysis, we see in the acceleration plots that there is a major swing in the Z axis in a sine wave like fashion, (displaying large, periodic oscillations in a sine wave like manner from around 4-18 m/s²) and we see a similar trend for the X and Y axis, albeit on a smaller scale due to my hand's motion not being perfectly "up and down" (there is some deviation in the side to side motion too) as expected and minimal forwards/backwards motion.

In the gyroscopic plots, we see a very consistent, clear cyclic pattern showing the peaks occurring when I change direction which is fully in line with what we expect. There are slight

variations due to my hands inducing a small pitch and roll movement, and my movement not being perfectly straight up and down.